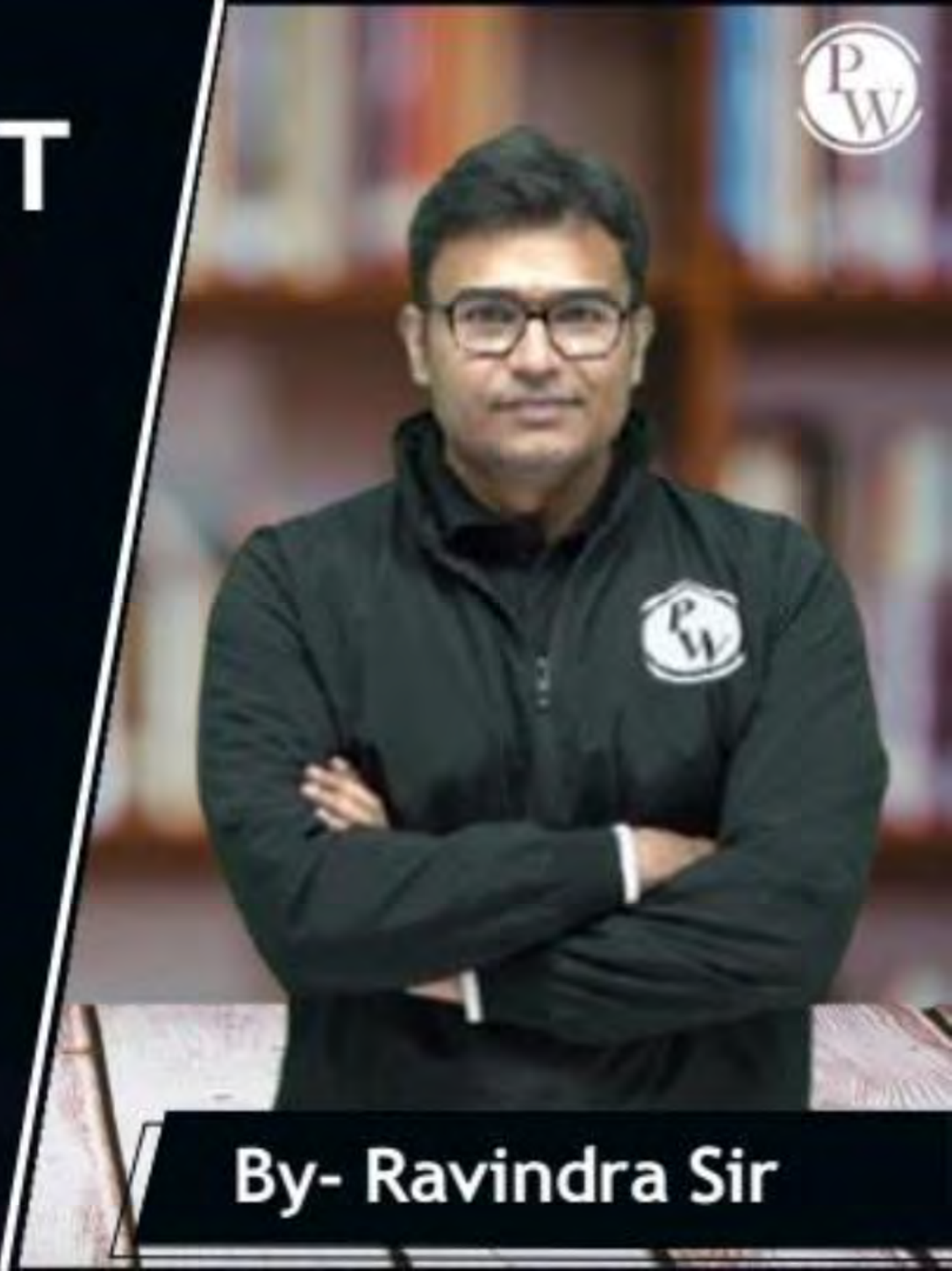
Computer Science & IT

COMPUTER NETWORK
(CN)

Hardware and Various Devices Used in Networking

Lecture No. 03



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

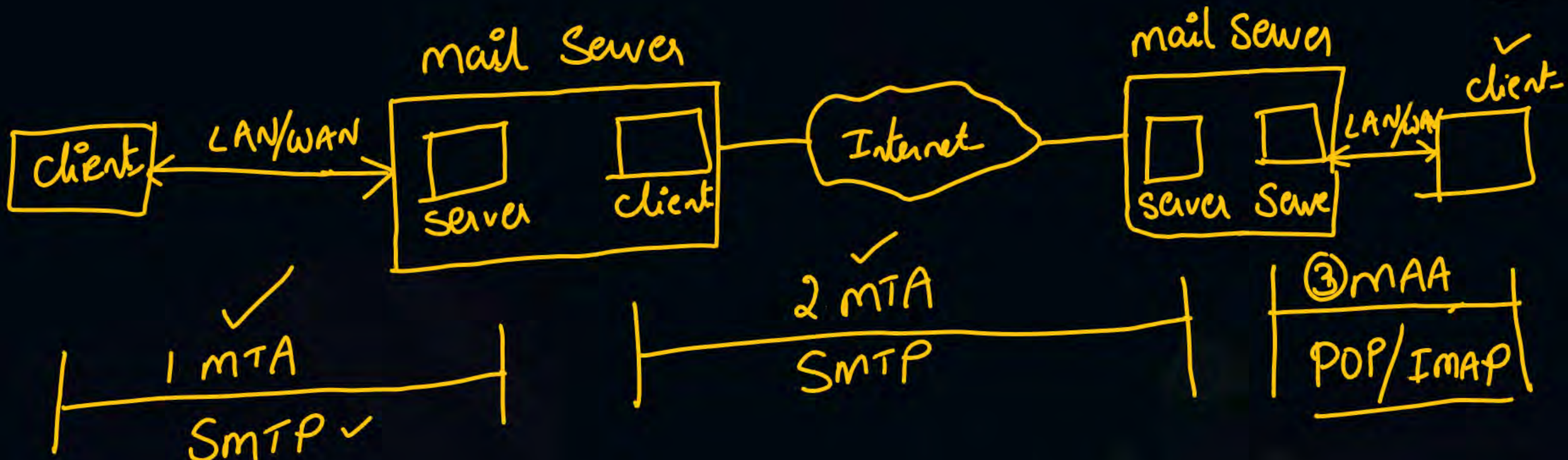
SMTP

Topics to be Covered



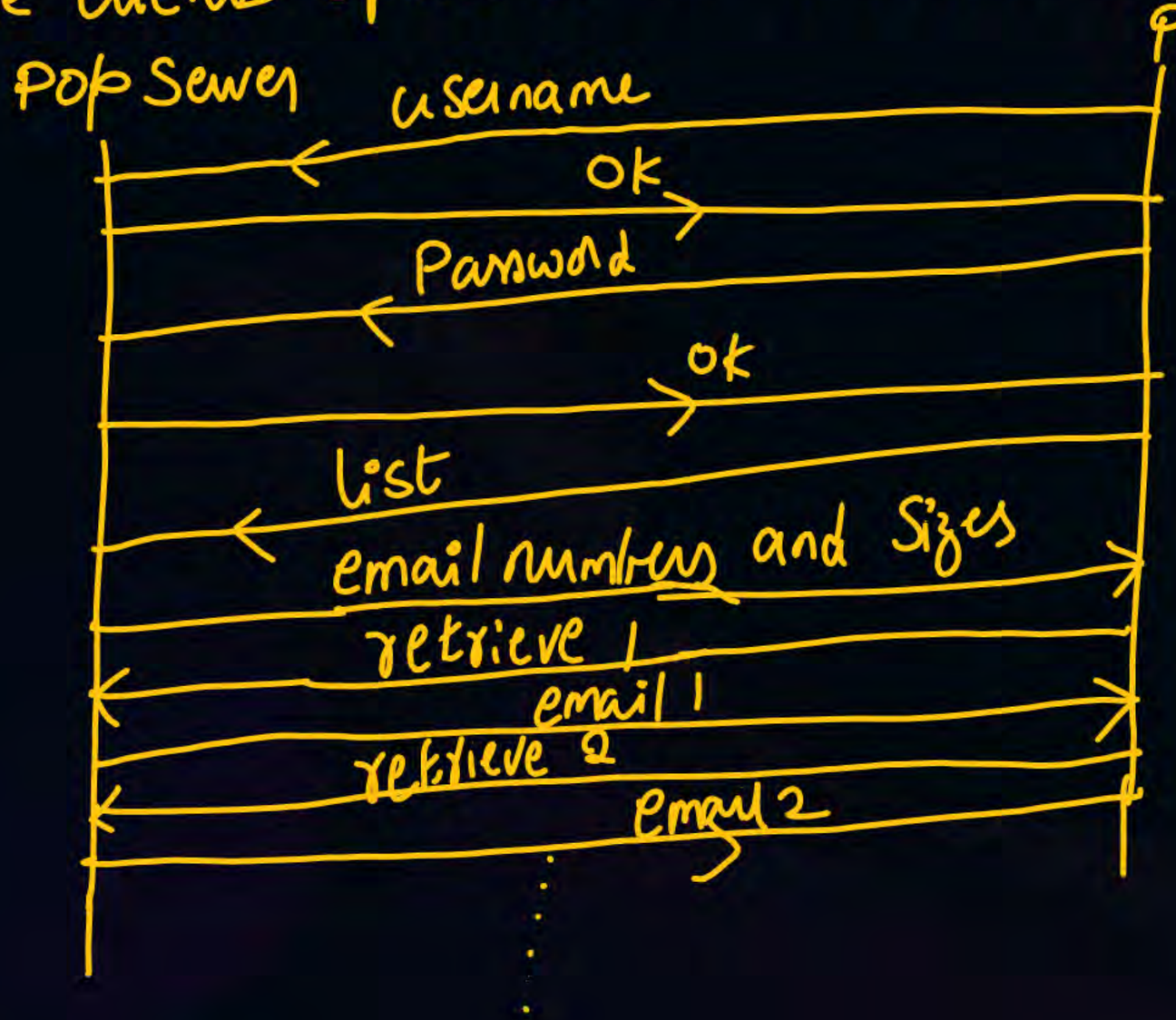
- Topic
- Topic
- Topic

POP, IMAP, FTP


mail Server
mail Server
client
Client
LAN/WAN
server
client
Internet
server
Server
LAN/WAN
client
1 MTA
SMTP
2 MTA
SMTP
3 MAA
POP/IMAP

POP3 :-

The client opens a TCP Connection with POP Server on port 110.

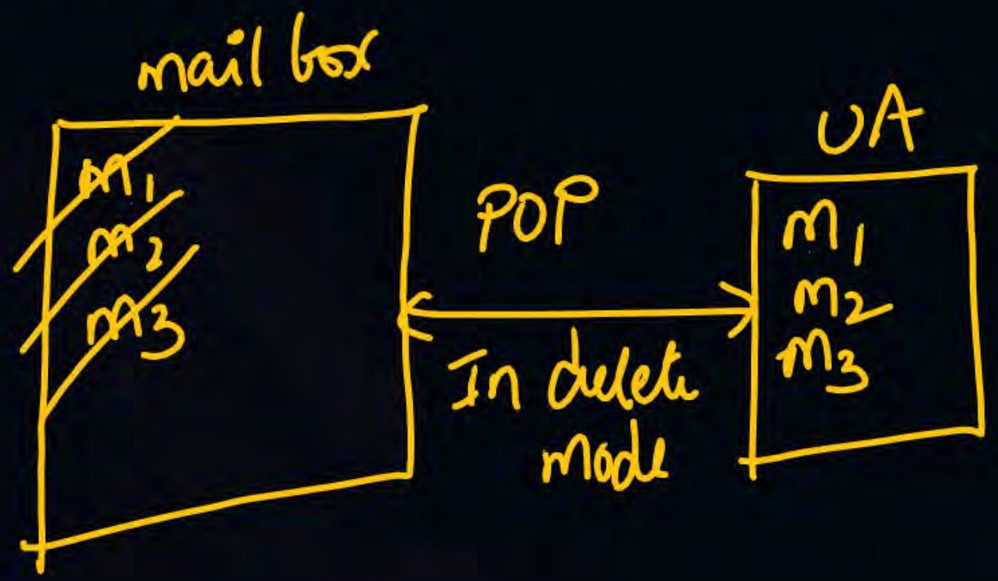


Pop has two modes :-

1. delete mode
2. Keep mode

Delete mode :-

In delete mode, the mail is deleted from mail box after each retrieval.



The delete mode is used when user is working at his permanent computer and can save and organise mail after reading or replying.

② Keep mode –

The keep mode is normally used when the user accesses the mail away from primary computer

The mail is read but kept in the inbox for later retrieval and organisation

IMAP4:-

IMAP4 is more powerful than POP

→ A user can check email header prior to downloading

→ A user can search the contents of email for specific string of characters prior to downloading

→ A user can partially download email

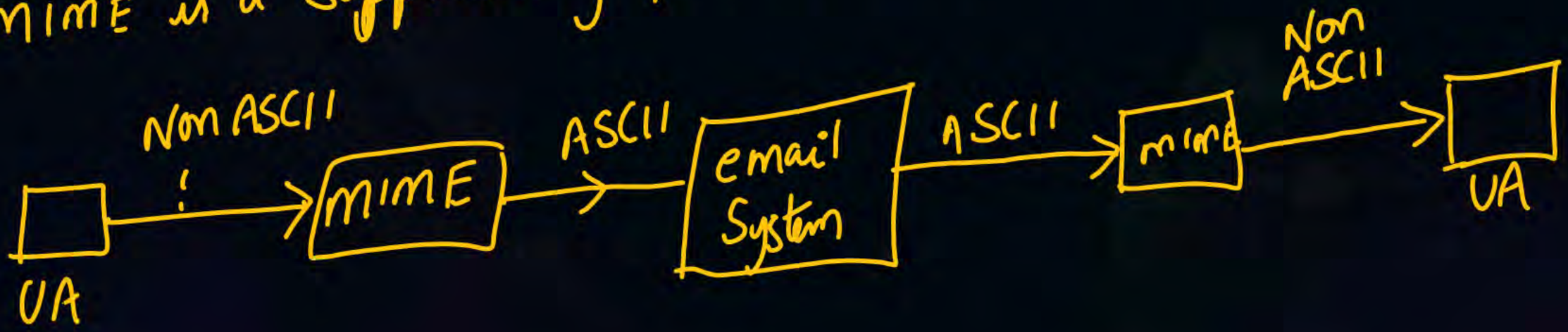
→ A user can create, delete & rename mail boxes on a mail server

→ A user can create a hierarchy of mailboxes in a folder for email store

MIME (multipurpose internet mail extension)

Electronic mail has a simple structure. It simply can send 7 bit ASCII format. It cannot send non english messages, & binary files & audio file & video files

MIME is a supplementary protocol that allows non ASCII format

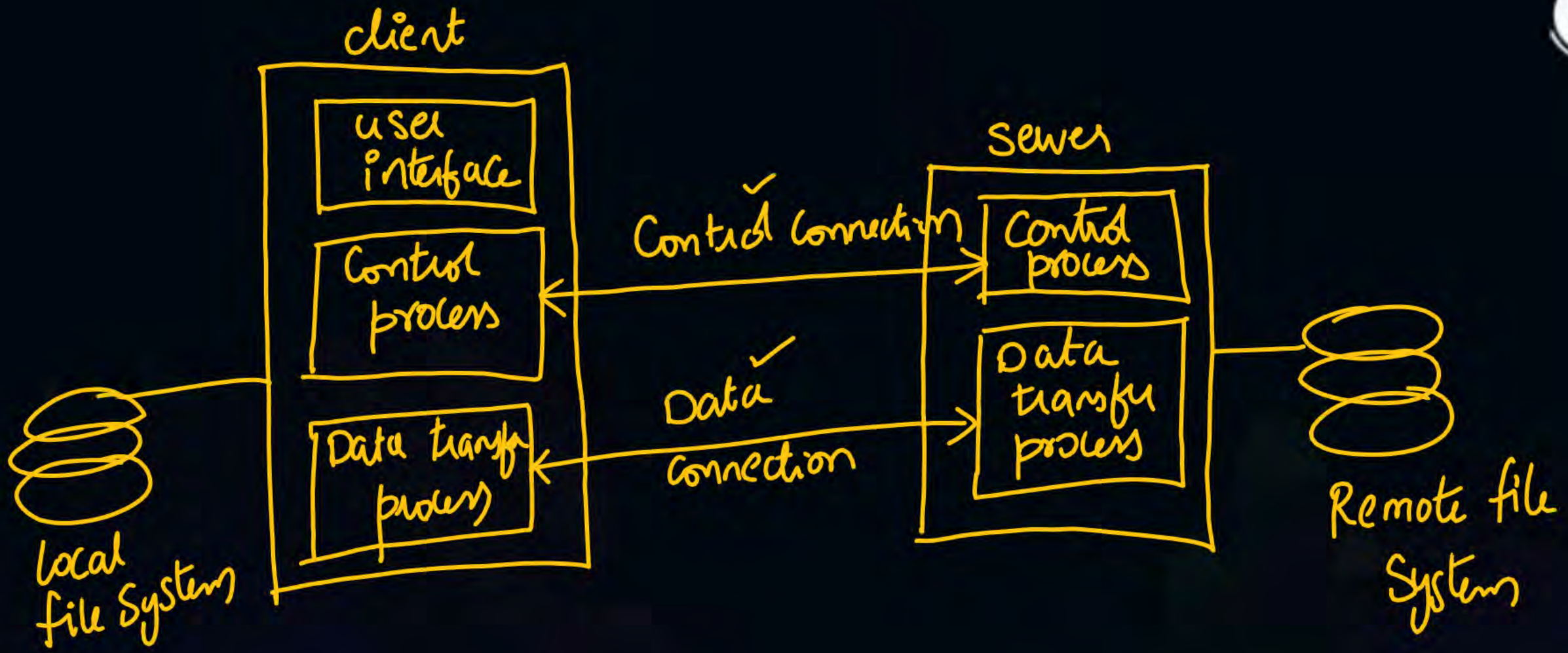


FTP (File transfer protocol)

FTP is a standard protocol by TCP/IP for copying a file from one host to another. Although transferring files from one system to another seems simple and straightforward, some problems must be dealt with. They are

1. Two systems may use different naming convention
2. Different ways to represent data
3. Different directory structures

Although we can use HTTP, FTP is a better choice to transfer large files & to transfer files using different formats


client
user interface
Control process
Data transfer process
Control connection
Data connection
Sever
Control process
Data transfer process
Local file System
Remote file System

Client has 3 components

1) user interface
2) client control process
3) client data transfer process

The server has two components :-

a) Server control process
b) Server data process

Commands are sent in Connection, Data is sent in data Connection. So it is called out of band protocol

Life time of the Connections :-

The Control Connection is persistant i.e it remain open during the entire interactive FTP session. The data connection is transient. The data connection is opened and then closed for each file transfer activity. It opens each time commands that involve transfering files are used, and it closes when file transfer is done

FTP uses TCP at port 21 for control connection and 20 for data connection

Control Connection:- Communication is achieved through commands and responses. This simple method is adequate for the control connection because we send one command (& response) at a time. Each line is terminated with two characters (carriage return and line feed).

FTP Commands:-

Command	arguments	Description
ABOR		abort the previous command
CDUP		Change the parent directory
CWD	Directory name	Change to another directory
DELE	File name	Delete a file
LIST	Directory name	lists sub-directories & files
MKD	directory name	Creates a new directory
PASS	user password	password

FTP Commands :-

Command	arguments	Description
PORT	port number	Client chooses a port
PWD		Display name of current directory
QUIT		logout of the System
RETR	File name	Retrieve a file
RMD	Directory name	Deletes a directory
RNFR	File name (old)	identifies file to be renamed
RNTO	File name (new)	Rename a file
STOR	File name	Store file; files are transferred from client to server

FTP Commans :-

Command	argument	description
STRU	F, R & P	Defines the data Organisation F: File R: records P: Pages
TYPE	A, E, I	Defaut file type (A: ASCII, E - EBCDIC, I: Image
USER	UserID	user information
MODE	S, B, & C	Define the transmission mode S: Stream B: Block C: Compressed

Response Code in FTP

125 – Data Connection open
150 – File Status okay
200 – Command okay
220 – Service ready
221 – Service closing
226 – closing data connection
230 – user login okay
250 – Request file action okay
331 – username is OK, password is needed

425 – Cannot open data connection
450 – File action not taken, file unavailable
452 – Action aborted, insufficient data
500 – Syntax error, unrecognised command
501 – Syntax error in parameters or arguments
530 – user not logged in

Data Connection :-

uses port 20 at Server site.

Creating a data connection is different from Control Connection

The following steps are followed

1. The client, not the server issues a 'open' using a port number. This must be done by the client because it is the client that issues the command for data transfer
2. The client sends its port number to the server using PORT Command

3. The server receives the port number and issues active open using well known port 20 and received port number

Communication over Data Connection:-

The client must define the type of the file to be transferred, the structure of the data, and transmission mode through control connection

Data's Structure:- FTP can transfer a file across data connection using on of the following interpretations of data

: a) File structure b) Record structure c) Page structure

a) The file structure format (used by default) has no structure

It is a continuous stream of bytes

b) In record structure, the file is divided into records

c) In page structure, the file is divided into pages having a page number and a page header.

File type :-

FTP can transfer one of the following types across the data connection

1) ASCII
2) EBCDIC
3) IMAGE

Transmission mode :-

FTP can transfer a file across the data connection using one of the following transmission modes

a) Stream of bytes

b) Blocks

File transfer

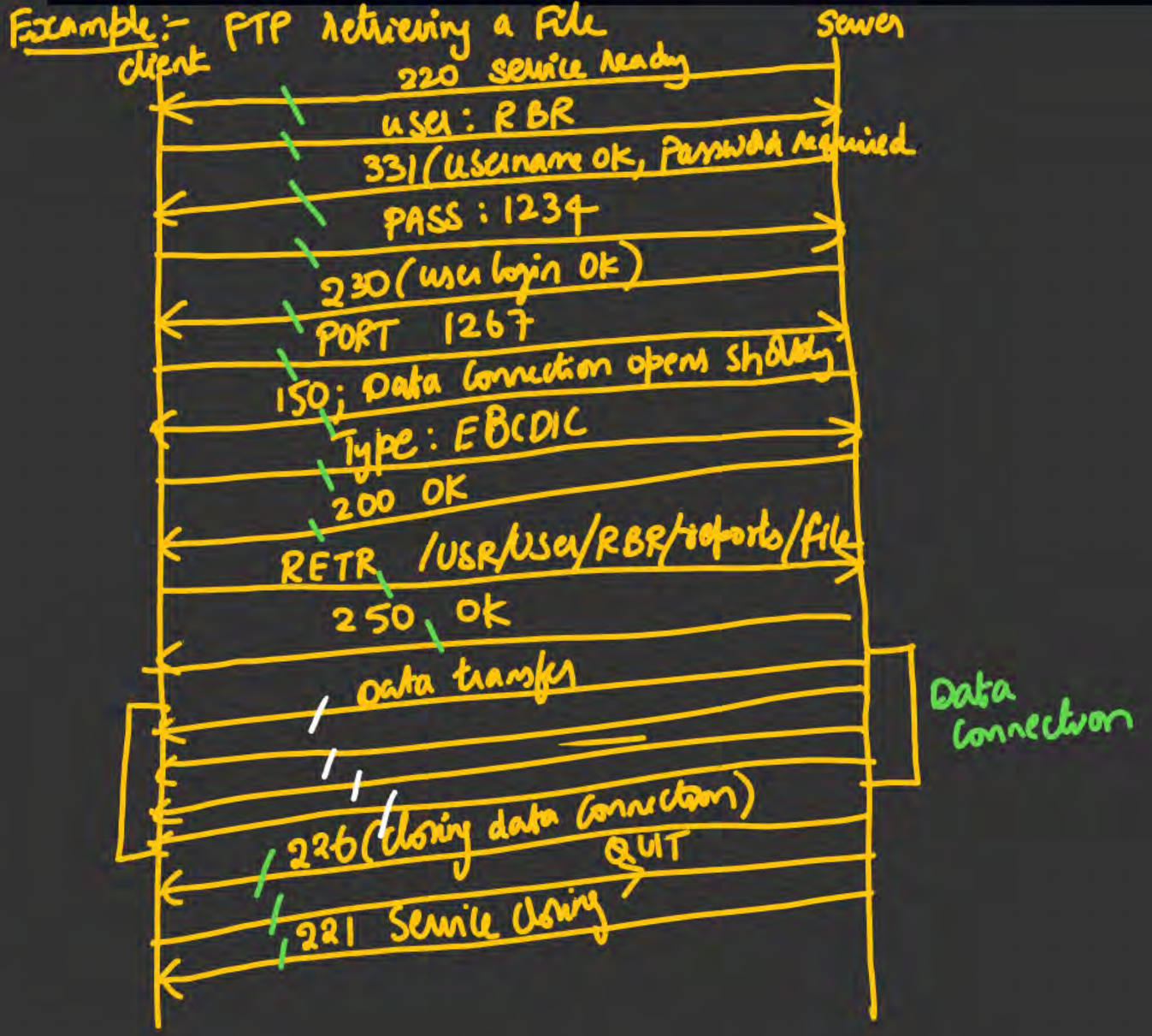
FTP transfer occurs over the data connection under the control of commands sent over control connection

File transfer in FTP means

a) Retrieving a file (Server to client)

b) Storing a file (client to server)

c) directory listing (Server to client)


Example:- FTP retrieving a file
Client
Server
220 service ready
user: RBR
331 (username OK, Password required
PASS: 1234
230 (user login OK)
PORT 1267
150; Data connection opens shortly
Type: EBCDIC
200 OK
RETR /usr/user/RBR/reports/file
250 OK
Data transfer
Data Connection
226 (closing data connection)
QUIT
221 Service closing

THANK - YOU